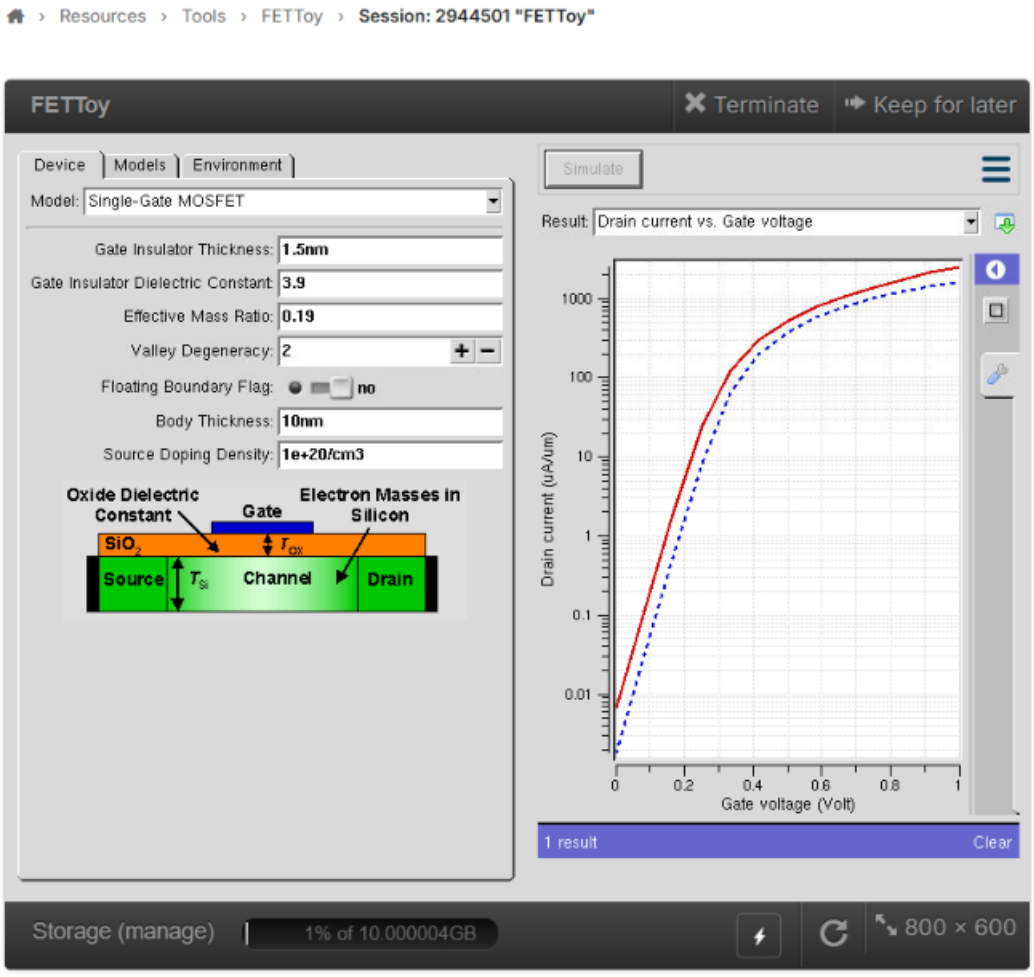
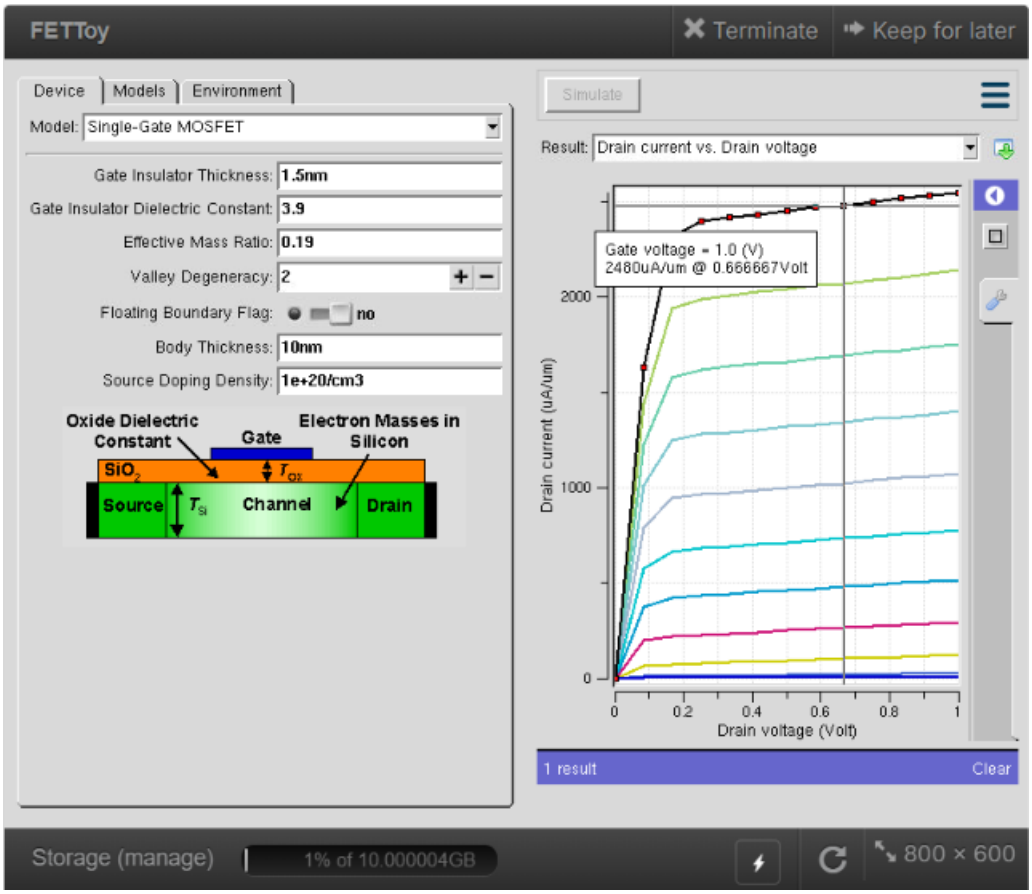
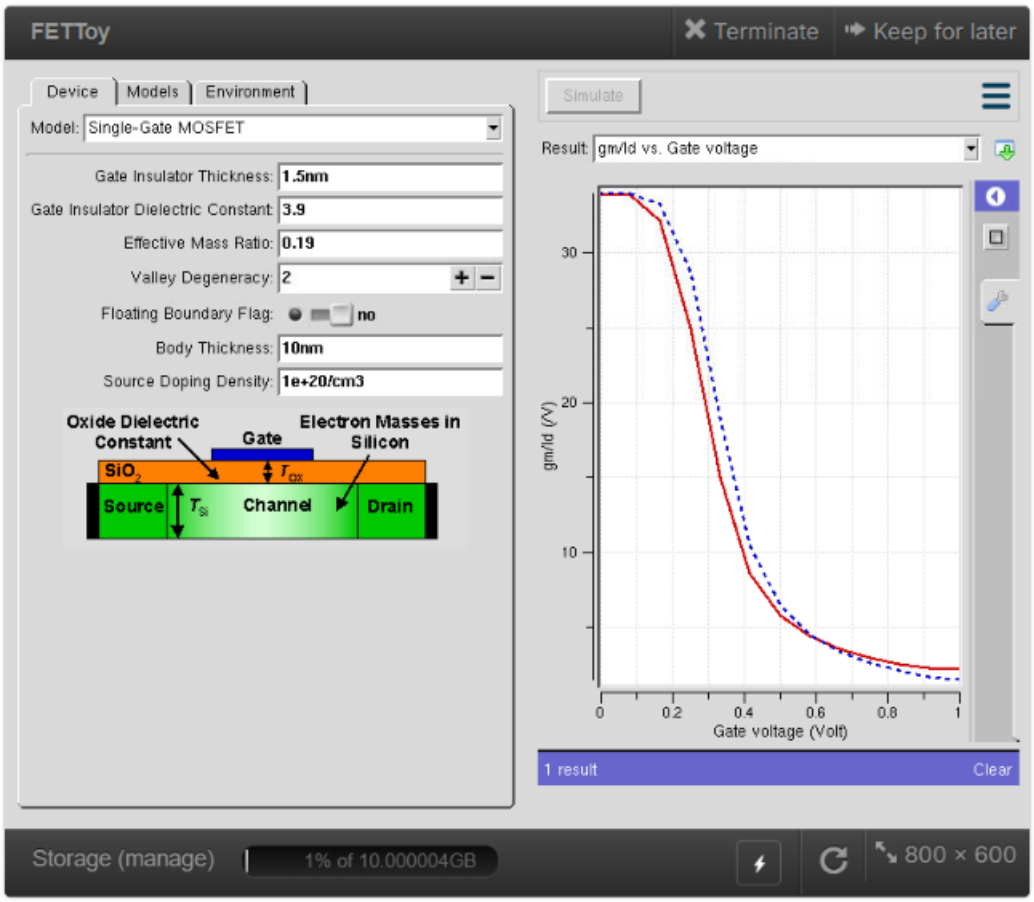
Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Objective:

To study the current-voltage characteristics and transconductance efficiency of a single-gate MOSFET using the NanoHUB FetToy Simulator.




FETToy
Terminate
Keep for later
Device
Models
Environment
Model: Single-Gate MOSFET
Gate Insulator Thickness: 1.5nm
Gate Insulator Dielectric Constant: 3.9
Effective Mass Ratio: 0.19
Valley Degeneracy: 2
Floating Boundary Flag: no
Body Thickness: 10nm
Source Doping Density: 1e+20/cm3
Oxide Dielectric Constant
Gate
Electron Masses in Silicon
SiO2
Source
Channel
Drain
Simulate
Result: Drain current vs. Drain voltage
Gate voltage = 1.0 (V)
2480uA/um @ 0.666667Volt
Drain current (uA/um)
Drain voltage (Volt)
1 result
Clear
Storage (manage)
1% of 10.000004GB
800 × 600



Reflection:

- **analyzed** the non-classical behavior of particles penetrating potential barriers using the Piece-Wise Constant Potential Barriers tool.
- **verified** the heavy exponential sensitivity of particle transmission and reflection probability relative to barrier thickness and height scaling.
- **learned** how quantum tunneling sets strict miniaturization limits in nanoelectronics due to subthreshold leakage currents in gate oxides.